Visualization – 15 Common Mistakes, 7 Tips, and 4 times to Break the Rules!

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This document summarises key points from my one-hour Data Visualization Masterclass, where you learn how to choose good visualizations designs, and analyse and present your data effectively.

Some of the examples are illustrated using the vizdataquality PyPI package that I developed (https://pypi.org/project/vizdataquality/) and others use MatPlotLib. However, the mistakes, tips and rule-breaking are relevant to all visualizations – irrespective of the software you use to create them.

Visualization masterclass structure

- Introduction
  - Why visualize data?
  - Anscombe's quartet
  - What chart type – how do I choose?
  - Data types
- Four well-known charts
  - Bar, pie, scatter & line
- Frequency distributions
  - Histogram, box, violin & datetime
- Visual encoding (including colour)
- Summary

Common mistakes

Mistake		Comment
1	Tiny font	For a solution, see Tip (a)
2	Hidden gaps in numerical axis	The example showed how the gaps would "pop out" if you include all of the years, not just those that had car parking fines
3	Truncated labels	For a solution, see Tip (b)
4	Illogical tick label order	E.g., order Y-axis labels starting from the top not the bottom
5	Invisible bars	For solutions, see Tip (c) and/or use perceptual discontinuity (see rule-breaking (i)). The vizdataquality bar chart Jupyter Notebook implements this.
6	Has overplotting mislead you?	It often does!
7	Are missing values handled correctly?	Are you interpolating values that are actually zero? Try drawing a line chart with gaps to make the missingness "pop out" (the vizdataquality line chart Jupyter Notebook implements this).
8	Do not plot categories on a line chart	But see rule-breaking (iii)
9	Pay attention to details	Always!

Mistake		Comment
10	What do a box plot's whiskers show?	In the caption, state what the whiskers show (there are a variety of different "standard" box plot conventions)
11	Assuming data has one mode	For a solution, see Tip (d) and the vizdataquality violin chart Jupyter Notebook
12	Patterns obscured by noise – check all levels of detail	Plot date/time frequency distributions at multiple levels of detail (e.g., year, month, day of the week, hour, minute and second). The vizdataquality datetime distributions at multiple levels of detail Jupyter Notebook implements this.
13	Legend and chart orders are different	Make them consistent!
14	Colours difficult to discriminate	For a solution, see Tip (e)
15	Not colourblind-safe	For a solution, see Tips (f) and (g). But see also rule-breaking (iv) if only you or your team will see the visualization

Tips

a) Make labels easy to read (use 18+ pt font for slides)
b) Use Y-axis if tick labels are long
c) Use labels to show small non-zero values
d) Violin plot reveals multi-modal distribution
e) Use a channel's full range
f) Consider alternatives to colour
g) Quick accessibility test – print in greyscale

Break the rules!

i. Distortion ("perceptual discontinuity") is good if it improves saliency
ii. Pie charts are sometimes clear with more than 6 wedges
iii. Use a line chart to show a distribution across categories
iv. OK if colours work for you!